

EXPERIMENT -2

Experiment Title:

Design and Simulation of 8×3 Encoder and 8×1 Multiplexer Using Logisim

Aim:

To design, implement, and simulate an 8×3 encoder and an 8×1 multiplexer using Logisim software, and to verify their truth tables and functionality through simulation.

Software Used:

Logisim (A graphical tool for designing and simulating digital logic circuits)

Theory:

Encoder (8×3)

An encoder is a combinational circuit that converts 2^n input lines into n output lines. An 8×3 encoder has: • 8 input lines (D_0 to D_7) • 3 output lines (A, B, C)

Only one input is active (high) at a time, and the output represents the binary code of the active input's position.

Truth Table:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A	B	C
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Boolean Expressions: • $A = D_4 + D_5 + D_6 + D_7$ • $B = D_2 + D_3 + D_6 + D_7$ • $C = D_1 + D_3 + D_5 + D_7$

Multiplexer (8×1)

A multiplexer (MUX) is a combinational circuit that selects one of several input signals and forwards it to a single output line. An 8×1 multiplexer has: • 8 data inputs (I_0 to I_7) • 3 select lines (S_1, S_2, S_3) • 1 output (Y)

The select lines determine which input is routed to the output.

Boolean Expression:

$$Y = \sum i = \sum (I_i \cdot m_i)$$

Truth Table:

S2	S1	S0	Selected Input	Output (Y)
0	0	0	I0	I0
0	0	1	I1	I1
0	1	0	I2	I2
0	1	1	I3	I3
1	0	0	I4	I4
1	0	1	I5	I5
1	1	0	I6	I6
1	1	1	I7	I7

Components Required

Component	Quantity
OR Gates	3 (for encoder)
AND Gates	8 (for MUX)
NOT Gates	3 (for MUX select inversion)
Input Pins	As required
Output Pins	As required
Wires	As required

Procedure:

8×3 Encoder:

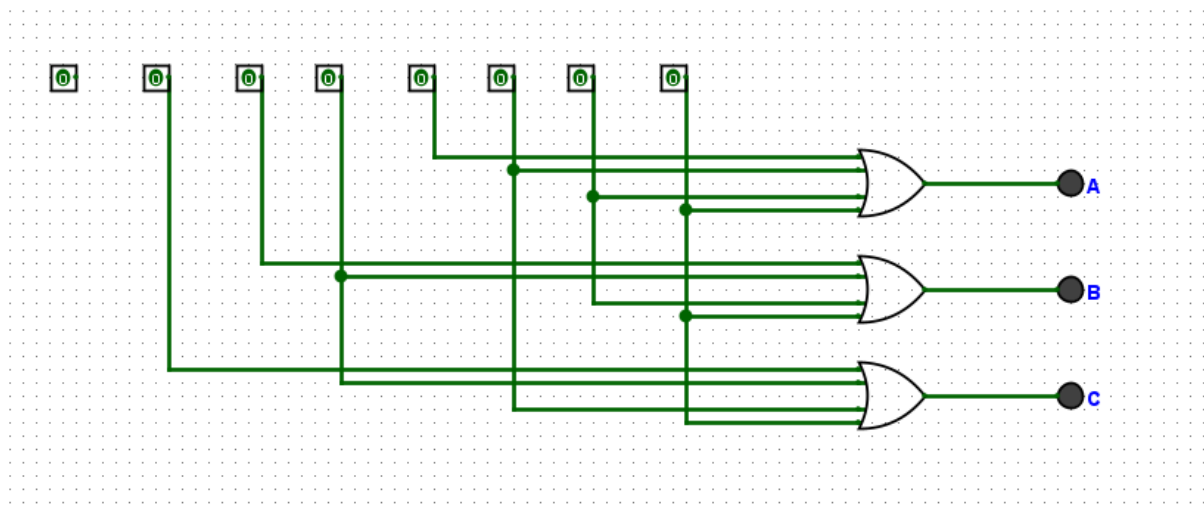
- Open Logisim and create a new project.
- Place 8 input pins labeled D_0 through D_7 .
- Place 3 output pins labeled A, B, and C.
- Connect inputs to OR gates according to the Boolean expressions.
- Wire OR gate outputs to the corresponding output pins.
- Simulate by setting one input HIGH at a time and verifying outputs.

8×1 Multiplexer:

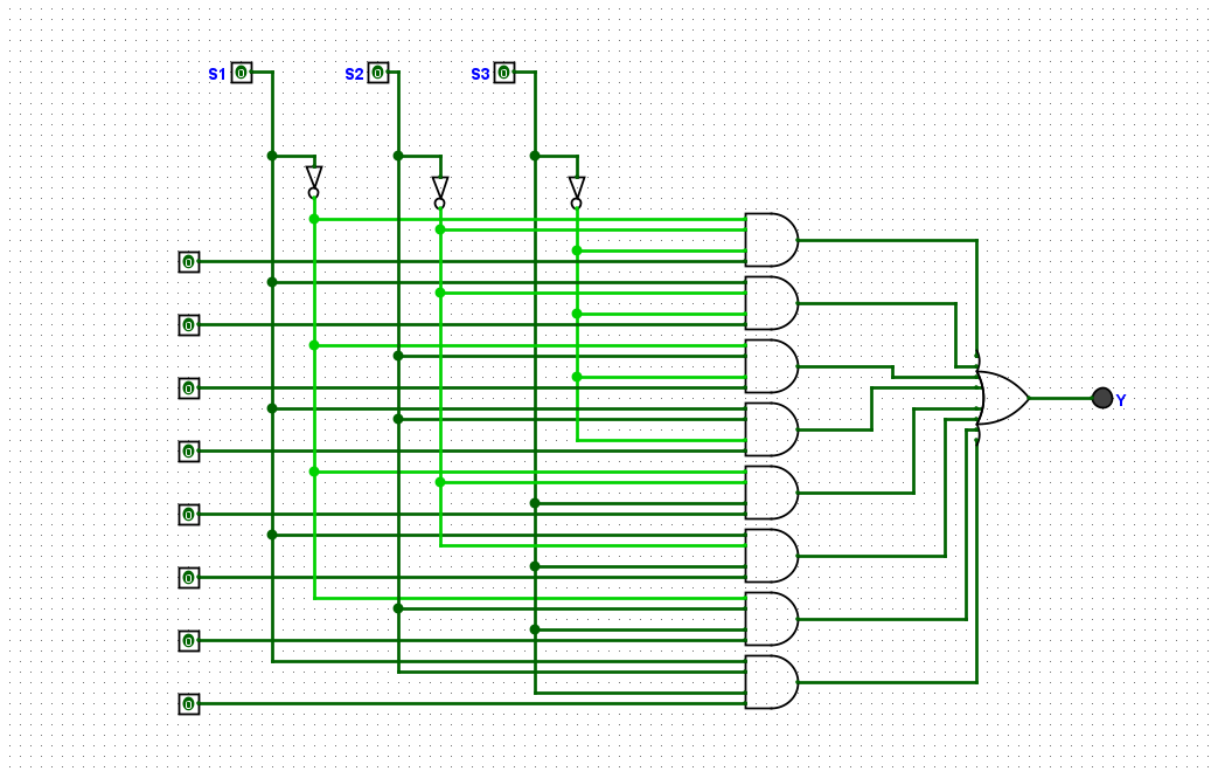
- Create a new circuit in the same project.
- Place 8 data input pins (I_0 – I_7) and 3 select input pins (S_1, S_2, S_3).
- Add NOT gates to generate complements of select lines.
- Place 8 AND gates (4-input each) — one for each data input.
- Connect each AND gate to the appropriate data input and select line combination.
- Connect all AND gate outputs to a single OR gate
- Wire the OR gate output to the output pin Y.
- Simulate by varying select lines and verifying output matches the selected input.

Circuit Diagrams:

Encoder (8×3)



Multiplexer (8×1)



Observations and Result:

Experiment	Input Condition	Expected Output	Observed Output	Status
Encoder	$D_3 = 1$, others = 0	A=0, B=1, C=1	A=0, B=1, C=1	Verified
MUX	S=011, $I_3=1$	Y=1	Y=1	Verified

Conclusion:

The 8×3 encoder and 8×1 multiplexer were successfully designed and simulated using Logisim. The simulation outputs matched the expected values from the truth tables, confirming correct circuit design. This experiment reinforced understanding of combinational logic design, Boolean algebra implementation, and digital circuit simulation.

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